

ZYRYANKA, Russian Federation

WMO: 254000

Lat: 65.7468N

Lon: 150.8870E

Elev: 141

StdP: 14.62

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-54.4	-51.6	-57.2	0.2	-54.1	-54.1	0.2	-51.3	15.6	-13.4	13.9	-15.1	0.9	100	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	82.0	66.2	77.6	63.7	73.6	61.4	67.8	79.1	65.1	75.0	62.7	72.1	6.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.5	88.1	74.2	60.9	80.2	70.9	58.3	73.1	68.3	32.3	79.6	30.2	75.0	28.4	72.2	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	14.6	12.8	-57.2	89.1	3.0	4.1	-59.3	92.0	-61.1	94.4	-62.7	96.6	-64.9	99.6
			-57.2	70.8	2.9	3.4	-59.3	73.2	-61.0	75.2	-62.6	77.1	-64.7	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	-32.9	-27.9	-9.9	13.4	38.9	56.7	60.9	54.3	40.9	15.1	-13.1	-28.8	(6)
(7)		DBStd	35.91	11.99	11.41	12.01	12.00	10.52	8.38	8.57	7.83	7.81	11.13	12.90	12.90	(7)
(8)		HDD50	13872	2570	2181	1856	1099	371	33	9	45	291	1081	1892	2444	(8)
(9)		HDD65	18629	3035	2601	2321	1549	811	267	185	340	723	1546	2342	2909	(9)
(10)		CDD50	806	0	0	0	0	26	235	348	179	18	0	0	0	(10)
(11)		CDD65	87	0	0	0	0	0	19	59	9	0	0	0	0	(11)
(12)		CDH74	833	0	0	0	0	3	199	545	85	1	0	0	0	(12)
(13)		CDH80	213	0	0	0	0	0	35	166	12	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	2.8	3.6	4.9	5.5	6.8	6.9	6.3	6.0	6.0	5.1	4.2	3.6	(14)
(15)	Precipitation	PrecAvg	11.5	0.7	0.6	0.4	0.3	0.6	1.3	2.0	1.7	1.3	0.9	0.9	0.7	(15)
(16)		PrecMax	19.0	2.0	1.7	1.2	1.5	1.7	3.1	7.0	4.3	4.8	4.4	2.2	1.6	(16)
(17)		PrecMin	5.7	0.1	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.0	0.1	0.2	0.1	(17)
(18)		PrecStd	3.1	0.4	0.4	0.3	0.3	0.4	0.7	1.3	1.0	1.0	0.7	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	2.6	27.9	46.0	72.6	84.2	89.8	81.0	70.0	42.6	20.6	6.0	(19)
MCWB			0.6	1.5	23.2	37.5	56.8	66.7	70.3	65.5	56.3	37.4	19.3	5.2	(20)	
2%		DB	-5.4	-3.4	22.0	41.3	66.1	79.5	84.5	75.9	62.6	35.4	13.8	-1.0	(21)	
		MCWB	-6.0	-4.0	18.7	34.4	52.8	62.9	68.3	62.8	51.8	31.5	13.1	-1.9	(22)	
5%		DB	-10.8	-7.2	15.9	37.3	60.8	75.3	80.1	72.1	57.8	32.0	8.4	-6.1	(23)	
		MCWB	-11.2	-7.7	13.6	31.6	49.7	60.5	65.9	60.7	49.6	30.0	7.6	-6.5	(24)	
10%		DB	-15.1	-10.7	8.7	32.9	55.9	71.2	75.6	67.8	53.0	29.5	4.2	-9.8	(25)	
		MCWB	-15.5	-11.1	7.2	28.6	46.6	58.4	63.9	58.8	47.0	27.6	3.5	-10.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	2.1	23.3	37.8	57.7	67.9	72.6	67.2	56.6	37.3	19.3	5.2	(27)
MCDWB			1.0	2.9	27.6	45.2	71.2	82.5	86.9	79.1	67.6	41.7	20.4	5.7	(28)	
2%		WB	-6.0	-4.0	18.8	34.6	53.5	64.2	69.7	64.1	53.0	32.2	13.0	-1.7	(29)	
		MCDWB	-5.5	-3.6	22.0	40.9	64.9	77.3	81.7	73.6	60.5	34.1	13.8	-1.0	(30)	
5%		WB	-11.1	-7.6	13.7	31.8	50.1	61.6	67.2	61.7	50.0	30.0	7.7	-6.6	(31)	
		MCDWB	-10.8	-7.2	16.0	37.0	59.8	73.1	77.3	69.9	56.9	32.1	8.4	-6.3	(32)	
10%		WB	-15.5	-11.1	7.2	28.7	46.8	59.2	64.8	59.4	47.3	27.6	3.5	-10.2	(33)	
		MCDWB	-15.2	-10.7	8.6	32.8	55.7	69.8	73.7	66.8	52.7	29.6	4.2	-9.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	14.3	21.2	21.9	16.5	17.1	16.1	15.1	12.3	10.2	10.7	10.4	(35)
MCDBR			11.9	12.3	27.9	21.4	23.0	23.6	23.9	22.9	22.3	12.4	11.5	12.3	(36)	
MCWBR			11.5	12.1	24.6	16.2	13.2	11.4	10.8	11.5	12.7	9.0	11.1	11.8	(37)	
5% WB		MCDBR	11.7	12.7	27.8	20.6	22.4	22.4	21.4	20.4	20.5	11.9	11.9	12.1	(38)	
		MCWBR	11.5	12.5	24.5	15.7	13.0	11.7	10.7	11.2	12.0	8.7	11.5	11.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.170	0.239	0.297	0.351	0.392	0.383	0.497	0.450	0.363	0.272	0.182	0.174	(40)	
(41)		taud	1.909	2.081	2.058	1.992	2.154	2.369	2.003	2.097	2.282	2.207	1.919	1.903	(41)	
(42)		Ebn at Noon	98	201	238	252	254	262	221	219	217	185	82	17	(42)	
(43)		Edh at Noon	6	18	30	42	40	33	47	39	25	16	5	2	(43)	
(44)	All-Sky Solar Radiation	RadAvg	34	254	821	1573	1821	1800	1563	1160	672	281	68	7	(44)	
(45)		RadStd	3	20	28	45	102	112	161	112	99	34	6	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.24	+2.12	+2.34	N/A	N/A	N/A	-542	-522	N/A	-41
(47)	Regional (1 neighbor)	+1.24	+2.12	+2.34	N/A	N/A	N/A	-542	-522	N/A	-41

Nomenclature: See separate page